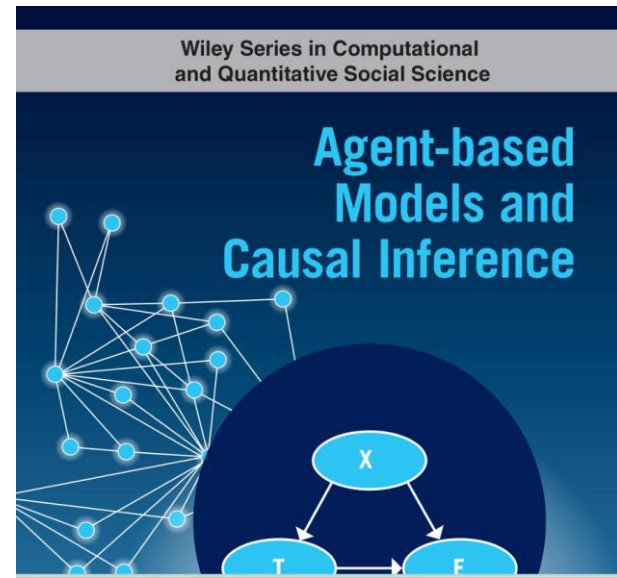


Motivation

- AI promises new possibilities for computational social science.
- Am a big fan of Gianluca Manzo who propounds evidential pluralism. Statistical inference + agent based models for proving causality
- Bro asks for KIDS(_{keep it descriptive stupid}) instead of KISS(_{keep it simple stupid}) models



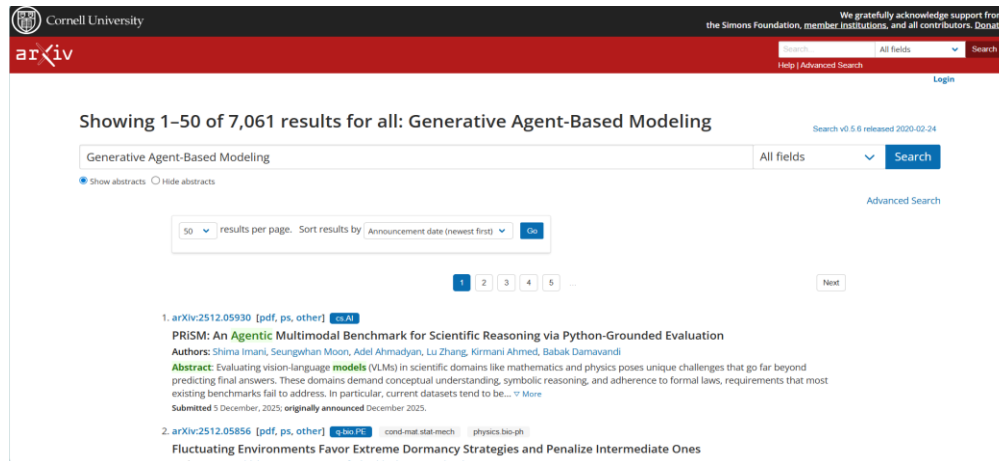
Problems with ABM

- Creating bespoke ABMs are too tedious and way beyond most academic people..(money, computational power, strong programming skills, granular data)
- In a way Manzo's vision is not very feasible at scale... I thought I could expedite ABMs with AI but not so.....u still need domain expertise

And much coveted empirical data which defeats the whole purpose because why bother ABM after running the elaborate RCTS or DiDs or IV regressions or whatever

Enter GABMs

- Generative Agent based models....super in vogue
- Almost every other week there is some paper....going to change the paradigm of social science research... think of the RCT revolution



- We are not academics and LLMs are stochastic, it's a black box, validation problems and so on. I don't have the knowledge or expertise to breakthrough academia. *But still want to revisit later to see how Manzo's generative theory of causality stands*

The Opportunity

- LLM agents now exhibit realistic cognition.
- This unlocks Synthetic Populations + Behavioral Simulation.
- Could make a case from empiric studies and papers published since 2023 and onwards(*I have ≥ 50 collated papers in my discord xD*)

Where we come in

- Forget academia ,causality and focus on enterprise
- Organizations rely on surveys, focus groups, and A/B tests.
- These methods are slow, expensive, biased, and cannot test counterfactuals
- Classic ABMs: unrealistic cognition. Need to hardcode properties. Object Based Programming is best for finding emergent behaviour after all but impossible to do in scale.
- LLM personas: no emergent behavior sure but can be leveraged otherwise
- Consultants: no simulation engine but people pay big bucks to McKinsey and so on
- Surveys: slow & expensive

Some ideas: synthetic consumer simulator

- **Persona Adoption:** LLMs are exceptionally good at role-playing. One can prompt an agent with a specific demographic profile
- RAG - Retrieval-Augmented Generation allow these agents to remember past interactions or hold specific brand biases, making them behave more like consistent individuals
- LangChain or AutoGen will allow us to spin up hundreds of these agents to interact with each other or a product simultaneously
- Many **strategic advantages** but **limitations** too... my gut feeling is in years or months to come this is literally commonplace

Viable is a spectrum. Technology is ready for early-stage validation and feels like converging. I mean it's the wild west



Some ideas: synthetic consumer simulator **cont.**

- There are open-source framework available right now for this exact purpose. Some are built specifically to handle the complex logic, memory retrieval, and component-based agent architecture needed to make synthetic consumers act like real people rather than just chatbots. *I will revisit this avenue later*
- Inference costs (API calls) are perhaps a question. But easy to do some back of napkin math to resolve" ***A synthetic consumer simulator is viable if the cost of the API tokens to simulate a consumer is lower than the cost of recruiting a real human."***

A7 ▾ 1. The Cost of a Real Human

[illegible]

Concordia, a library developed by Google DeepMind for constructing and managing **Generative Agent-Based Models (GABMs)**.

- **How Concordia Works**
- **The Game Master (GM):** Inspired by tabletop role-playing games (like Dungeons & Dragons), Concordia uses a "Game Master" agent. The GM is responsible for simulating the environment, describing the effects of actions, and maintaining the "rules" of the world (physical, digital, or social).
- **Generative Agents:** Agents execute actions by describing what they want to do in natural language. The GM translates these descriptions into implementations, such as checking physical plausibility or making API calls to real digital apps.
- **Component System:** Agents are built using a modular "component" system. These components (e.g., identity, plans, observations, memory) mediate between the agent's long-term memory and the LLM to produce behaviour.

Google DeepMind Releases Concordia Library v2.0

Google DeepMind has released Concordia v2.0, a platform providing a robust environment for creating multi-agent simulations, games, and AI situations.

The screenshot shows the GitHub repository for Concordia. At the top, there's a search bar and navigation links for Actions, Security, and Insights. Below that, statistics show 22 watchers, 239 forks, and 1.1k stars. The repository has 24 tags and a recent commit 'Minor: Fix link path' by f5c4007. A table of recent commits is visible, including updates to Python requirements, pull request merges, and documentation changes. On the right, the 'About' section describes it as a library for generative social simulation, with tags for multi-agent, social-simulation, agent-based-simulation, and generative-agents. It also lists the README, Apache-2.0 license, and contributing guidelines.

Commit Message	Time Ago
Require Python >= 3.12	2 weeks ago
Merge pull request #170 from google-deepmind:dependab...	last week
[vscode] only organize imports on explicit save	last year
Remove --all-extras from pyproject.toml to give more flexibil...	5 hours ago
move prefix_entity_name to config parameters	2 hours ago
Add Concordia cheat sheet.	4 days ago
Copybara import of the project:	last year
Update pylintrc from canonical link	last week
Require Python >= 3.12	2 weeks ago
Increment version number to 2.1.0	4 months ago
update citation bib	2 years ago

Why lean towards this open-source library rather than ground up

- [2312.03664](#)
- **Logic of Appropriateness:** Instead of optimizing for a specific reward (like in Reinforcement Learning), Concordia agents act based on what a person with their specific identity and situation *would* do.
- **Not Rational Maximizers:** The agents are not modeled as "Homo economicus" (rational utility maximizers). They do not have an explicit utility function but instead rely on the LLM's training on human culture to simulate behavior.
- **Social Construction:** The model supports theories where agents and social structures (norms, rules) co-constitute one another, allowing for the simulation of complex social phenomena.

Use cases

- In the paper the authors cite several applications like:
Digital User Studies: The library can simulate users interacting with digital technologies (like smartphones and apps) to evaluate services or generate synthetic user data without privacy concerns.
- **Social Science Research:** simulate elections, small business interactions, or resource management scenarios (*no joke they mention fisheries*) to study social dilemmas and policy effects.
- **Psychological Modeling:** Researchers can implement and test classic psychological theories, such as the "theory of planned behavior," by creating components that represent attitudes and norms.
- **Auditing AI:** The system allows for transparent auditing. Because agents "think" and plan in natural language, their "chain of thought" can be reviewed to understand why they made specific decisions.

How It Works in my minds eye

- 1. Define population.
- 2. Define scenario.
- 3. Run simulations.
- 4. Analyze emergent outcomes with dashboards. *This part is important.... People are clamoring for a platform that facilitates and abstracts away the coding and API calls*

At any rate one must tinker and prototype and ideas/inspiration will follow

Hmm, well well

US!!!

5 Conclusions and Future Work

In conclusion, we have provided a guide for designing experiments using generative agent-based models within the Concordia framework, highlighting the benefits of this modeling for simulating complex environments and offering a structured approach to tool selection and model design.

Future work could focus on making GABMs even more accessible by setting up an interactive web platform, allowing users to create and customize characters, adjust their behaviors, and visualize detailed information about the agents. It could also offer simulations of interactions between agents with similar or contrasting personalities or political ideologies, ultimately broadening their applicability and impact across various fields.

Differentiation

- Pivot to Italia... maybe even Emilia Romagna enterprise
- Not a chatbot. Not a classic ABM.
- Cooperatives generate ~ **30-40% of the region's GDP**.
We are talking about giants like Coop Alleanza 3.0, Unipol, and SACMI and so on. Unlike the US or UK, which rely on giant vertical corporations, this region relies on networks of small-to-medium enterprises (SMEs) and **cooperatives** that collaborate with each other
much to explore....
- *Even comune di bologna....for ex-ante policy analysis...they even open source a lot of their data*

Sights here

- Marketing & product testing
- Policy simulation
- Organizational digital twins
- Crisis forecasting

Come up with demo examples and must include prototype before sending application!

Technical architecture

Simulation Engine

- Concordia (multi-agent cognition)
- Python orchestrator (behaviors, events, scoring, environments)
- Validation module (Monte Carlo runs) / *saw another interesting architecture elsewhere with ExpectedParrot*

Data Layer

- client-uploaded data → preprocess → encode into agent priors *abstract away their coding*
- regulatory-friendly storage *must be mindful of EU regulation though idk*
- embeddings of qualitative data

Agent Generation

- persona templates
- demographic distributions
- domain embeddings
- LLM memory architecture (Concordia)

- goal generation from client prompts

Scenario Engine

- interventions (price change, new message, layout, policy)
- environment constraints
- simulation horizon
- parallel runs

Dashboard Layer

- Plotly/React-based
- multi-run comparison
- “explainable AI agent logs”
- emergent network patterns

I think there needs to be something novel about the interface... even frontier labs is on cusp of overturning chatbot GUI

API Layer

- SaaS connector
- export results to PDF/CSV

Business Model

- Not an expert and not much knowledge will think and expound later.....need not bother now

Roadmap for 1 year

- Q1: MVP + pilots *getting the most juice out of ALMACUBE by this time lol*
- Q2: dashboards + first clients
- Q3: modules R&D
- Q4: enterprise + government